

Another idea of an alternative cyborg called Lobster is one fabricated not by using internal implants but by using an external shell such as a powered exoskeleton. It is the reverse of human cyborgs that appear human externally but are synthetic internally. Lobsters look inhuman externally but are human internally.

In the movie *Batman: The Dark Knight Returns Part 2*, Batman puts on a powered exoskeleton for his combat with Superman. This enables him to lift the Batmobile easily with one hand and fight on equal terms with Superman.

The US Army is in the process of creating an Iron Man-like suit that imparts super-human strength to the user. This exoskeleton is called Tactical Assault Light Operator Suit (TALOS) and will have a strata of smartmaterials built-in with sensors, as well as a wearable computer very much like Google Glass. It will be able to keep an eye on soldiers' critical parameters and multiply the available strength using hydraulics.

Powered exoskeletons also enable firemen and other rescue workers to endure hazardous conditions. In the healthcare sector, their application ranges from carrying out very accurate surgeries to enabling nurses to shift bulky patients. And then there are several other important uses for people with disabilities and special needs.

ReWalk is a commercial exoskeleton and a bionic walking-assistance device that uses powered leg attachments to enable paraplegics to stand straight, walk and climb steps.

Then, we have animal cyborgs. RoboRoach is the result of a project started at University of Michigan, USA, by a biomedical engineering student. It is a kit that enables



Fig. 2: Illustration of TALOS to be used by the US Army (Image courtesy: sci-techuniverse.blogspot.com)

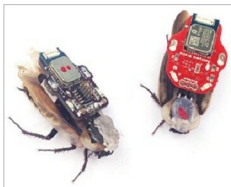


Fig. 3: RoboRoaches, the animal cyborgs (Image courtesy: www.extremetech.com)



Fig. 4: A man wearing a specialised camera for an eye (Image courtesy: s.telegraph.co.uk)

students to use micro-stimulation to fleetingly control the movements of a walking cockroach (in left and right directions) using a Bluetooth-enabled smartphone as the controller.

Cyborg applications

In the field of medicine. In the medical field, broadly there are two types of cyborgs: restorative and enhanced. Restorative technologies help restore lost function, organs and limbs by repairing broken or missing process-

es to a healthy or average level of function.

On the other hand, an enhanced cyborg aims to surpass normal processes or even add new functions that were not originally there. Bionic implants enable model organs or body parts to impersonate the original function and often even surpass these.

A brain-computer interface, or BCI, provides a direct path of communication from the brain to an external gadget, success-

fully generating a cyborg.

Retinal implants are another practice of cyborgisation in medicine. Efforts are on towards perfecting this technology, and there have been major advances in the use of electronic stimulation of the retina to enable the eye to sense patterns of light. A specialised camera is worn by the user on the frames of their spectacles, which converts the image into a pattern of electrical stimulation.

A chip present in the user's eye electrically stimulates the retina with this pattern by exciting certain nerve endings that transmit the image to the optic centres of the brain and the image is seen by the user. Such type of innovation could enable blind people to see again.

Similarly, mute people could start speaking again. A surgery can enable redirection of the nerve that controls the voice and sound production to a muscle in the neck. A sensor picks up electrical signals, which are then transmitted to a processor to control the timing and pitch of a voice simulator. The simulator, in turn, vibrates, producing a multi-toned sound that can be shaped into words by the mouth.

With advances in medical technology, developments such as implantable silicon silk electronics, augmented reality and QR codes are narrowing the gap between tech-